Lab Experiment 8: Conduct Z test for the Data Given using Python Objective:

Aim:

To test whether the average weight of a species of birds differs from 150 grams.

Procedure:

1. Null Hypothesis (H0): The average weight of the birds is 150 grams.

2. Alternative Hypothesis (H1): The average weight of the birds is not 150 grams.

3. Sample: Measure the weights of 30 birds randomly selected from the population.

4. Z-Test: Conduct a Z-test to compare the sample mean to 150 grams.

5. Decision Rule: Use a significance level of α = 0.05.

Python Code Example:

```
import numpy as np

import scipy.stats as stats

sample_data =
np.array([152,148,151,149,147,153,150,148,152,149,151,150,149,152,151,148,150,152,149,150,148,153,151,150,149,152,148,151,150,153])

population_mean =150

sample_mean = np.mean(sample_data)

sample_std = np.std(sample_data,ddof=1)

n=len(sample_data)

z_statistic = (sample_mean - population_mean)/(sample_std/np.sqrt(n))

p_value = 2*(1-stats.norm.cdf(np.abs(z_statistic)))

print(f'Sample Mean: {sample_mean:.2f}')

print(f'z_statistic: {z_statistic:.4f}')

print(f'P-Value: {p_value:.4f}')

alpha=0.05

if p_value<alpha:

  print('Reject the null hypothesis: The average weight is significantly different from 150 grams.')

else:

  print('Fail to reject the null hypothesis: There is no significant difference in average weight from 150 grams.')
```

Output:

Sample Mean: 150.20

z_statistic: 0.6406

P-Value: 0.5218

Fail to reject the null hypothesis: There is no significant difference in average weight from 150 grams.

Result:

Fail to reject the null hypothesis: There is no significant difference in average weight from 150 grams.